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Effects of multi-herb and ashwagandha root formulas on stress modulation: a randomized, double-blind, placebo-controlled clinical study

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Abstract

Background Chronic stress is detrimental to the maintenance of the main response system — the hypothalamic–pituitary–adrenal (HPA) axis. The current study aimed to investigate the efficacy of two plant-based adaptogens, a formula containing *Rhodiola*, holy basil and *Schisandra chinensis* (VL-G-A57) and a full-spectrum ashwagandha (VL-G-E12), on stress and related symptoms in individuals with high stress.

Materials and methods The 60-day randomized, double-blind, placebo-controlled clinical study included individuals aged between 18 to 65 years with a body mass index (BMI) of 18 to 29.9 kg/m². One hundred eighty-six participants were randomized to one of the adaptogens, VL-G-A57 or VL-G-E12, or to placebo. The primary outcome was a reduction in stress levels. Secondary outcomes were changes in sleep quality, fatigue, restorative sleep, mental alertness, mood dysregulation, and anxiety. A priori power analysis determined the required sample size. Efficacy was assessed by comparing mean changes in the primary endpoint at days 30 and 60 using ANCOVA, with base-line values as covariates. Dunnett's post hoc test identified significant differences versus placebo, and within-group changes were evaluated using paired *t*-tests. Normality was assessed visually and via Shapiro–Wilk/Kolmogorov–Smirnov tests as needed. Secondary outcomes were analyzed similarly. Analyses were conducted using R (v4.0.5) and XLSTAT (v2021.3.1).

Results At day 60, both VL-G-A57 and VL-G-E12 significantly reduced Perceived Stress Scale (PSS) scores compared to placebo ($p < 0.0001$). Sleep quality, as measured by the Pittsburgh Sleep Quality Index (PSQI), improved significantly in both adaptogen groups (VL-G-A57: $p = 0.0008$, VL-G-E12: $p < 0.0001$). This corresponded well with the Restorative Sleep Questionnaire–Weekly (RSQ-W) results in the two IP arms when compared with placebo ($p < 0.0001$). Additionally, mood dysregulation (VL-G-A57: $p = 0.0454$), anxiety (VL-G-A57: $p = 0.0004$, VL-G-E12: $p = 0.0015$), and stress levels (VL-G-A57–VL-G-E12: $p < 0.0001$) showed significant improvements compared to placebo. No differences in mental alertness were observed.

Conclusion The study concluded that both VL-G-A57 and VL-G-E12 were associated with reductions in stress, fatigue, and anxiety while improving mood and sleep quality.

Trial registration ClinicalTrials.gov NCT05602389 and the Clinical Trials Registry — India CTRI/2022/11/047635. Registered on 1 November 2022 and 24 November 2022

Keywords *Rhodiola rosea*, Ashwagandha, Stress, Anxiety, Sleep quality, Fatigue

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Introduction

In today's fast-paced lives, stress is a common phenomenon; however, chronic stress affects overall well-being and requires active intervention. Studies show that chronic stress is detrimental to mental and physical health [1]. Adaptogens are known to reduce reactivity to environmental stressors and contribute to the healthy regulation of stress hormones by supporting the body's physiological stress response [2, 3]. In recent years, an increasing number of adaptogens have been launched onto the global market. According to a market-research report, the US adaptogens market is predicted to grow at a compound annual growth rate of 7.2% from 2025 to 2030 [4].

Adaptogens were initially defined as substances that enhance the “state of nonspecific resistance” during stress. Studies on animals and isolated neuronal cells have revealed that adaptogens exhibit neuroprotective, anti-fatigue, antidepressive, anxiolytic, nootropic, and central nervous system (CNS)-stimulating activity. Several clinical trials demonstrate the anti-fatigue effect of adaptogens, increasing mental work capacity. Recent meta-analyses of several adaptogens have provided a rationale for these effects at the molecular level. It was concluded that the mechanism of action of adaptogens is associated with the regulation of the HPA axis, leading to reduced levels of cortisol and subsequently modulating the key mediator of stress response [5–7].

The objective of the current study was to assess the adaptogenic effects of two investigational products (IPs): Adrenal Health Daily Support/Daytime HPA (VL-G-A57), a formula containing *Rhodiola*, holy basil, milky oats extract, ashwagandha, and schisandra and ashwagandha roots (VL-G-E12). Both preparations are manufactured by Gaia Herbs, Brevard, North Carolina, USA, and utilize full-spectrum extracts, with VL-G-A57 additionally including an oat preparation in the American Eclectic tradition, using fresh seed pods in the “milky” stage (green with a milky or jelly-like endosperm) [8].

Ashwagandha (*Withania somnifera* (L.) Dunal) has been used for years to enhance positive mood states, deepen sleep, and improve memory [5, 9, 10]. *Rhodiola rosea* L., or “golden root,” has garnered attention as an adaptogen for its role in improving depression, enhancing work performance, eliminating fatigue, and treating symptoms of asthenia [11]. Holy basil (*Ocimum sanctum*) is a potent adaptogen with antimicrobial, anti-inflammatory, antihypertensive, and immunomodulatory effects [12–16]. Oats (*Avena sativa* L.) have traditionally been used as a demulcent, diuretic, and antispasmodic in treating insomnia and as a nervous system tonic in the milky phase [8, 17]. *Schisandra chinensis* (Trucz.) is

another traditional Chinese medicine with antioxidative, neuroprotective, anticancer, vasorelaxant, and cytoprotective properties among the most studied ones [18–20]. Collectively, the formulation provides a balanced and complementary blend of adaptogens and nervines, delivering a diverse phytochemical profile designed to act on stress responses and mediate its impacts on bodily functions (inflammation, oxidative stress, liver and nervous system function).

The primary objective of the present study was to explore the effect of the 60-day administration of the two IPs on perceived stress. The secondary outcomes included the impact of the IPs on sleep quality, fatigue, restorative sleep, mental alertness, depression, and anxiety.

Methods

Study design

The study was designed as a randomized, double-blind, placebo-controlled clinical trial to investigate the effect of 60-day administration of two investigational products — VL-G-A57, a proprietary blend of *Rhodiola rosea* root, holy basil leaf, milky oat seed, *Schisandra* berry, and ashwagandha root, and VL-G-E12, ashwagandha root, a proven adaptogen, compared to the placebo on stress in highly stressed individuals. The study was conducted at five clinical sites in Mumbai, India, under the supervision of qualified physicians from December 8, 2022, to July 12, 2023.

Participants

The study was conducted among adults aged between 18 to 65 years with a BMI range of 18 to 29.9 kg/m² and moderate levels of physical activity as per the International Physical Activity Questionnaire–Short Form (IPAQ–SF). Highly stressed individuals with a PSS score in the range of 27–40 and an RSQ-W score of ≤ 50 were included in the study. Individuals with the following criteria were excluded from the study: (1) medical history of heart disease, respiratory disorders, metabolic or lifestyle disorders, seizure disorders, or other chronic health conditions requiring medication; (2) a mental-health disorder as assessed by Mini-International Neuropsychiatric Interview 5.0.0 [21] or taking psychotropic medications; (3) blood pressure range of systolic ≥ 140 mm Hg and diastolic ≥ 90 mm Hg; (4) history of severe intrinsic sleep-related disorders; (5) usage of sleep medication, sleep supplements, and/or ashwagandha and/or other supplements in the last 3 months; (6) known hypersensitivity to ashwagandha; (7) working a night shift; and (8) pregnant or nursing women. All participants were instructed not to make any major lifestyle changes during the study period.

The prospective participants fulfilling the inclusion–exclusion criteria were allocated to one of the three study groups, VL-G-A57, VL-G-E12, or placebo, in a 1:1:1 ratio at the randomization visit following a computer-generated random sequence using a block randomization method with a block size of six. The randomization chart was generated by an independent statistician using Stats-Direct software (Ver. 3.1.17). The participants, research staff, and investigators were blinded to the study product allocation.

Intervention

The investigational product labeled VL-G-A57 comprises a distinctive proprietary blend of *Rhodiola rosea* root, holy basil (*Ocimum sanctum* L.) leaf, oat (*Avena sativa* L.) milky seed, *Schisandra chinensis* (Trucz.) berry, and ashwagandha (*Withania somnifera* (L.) Dunal) root standardized by high-performance liquid chromatography to deliver 4–6 mg of eugenol and 6 mg of total rosavins per serving. The investigational product, VL-G-E12, is an adaptogenic supplement of ashwagandha hydroethanolic extract and ashwagandha root powdered extract standardized by high-performance liquid chromatography to deliver no less than 2.5 mg of withanolides per serving. The detailed composition of the investigational products is given in Table 1. According to previous clinical studies, the use of ashwagandha root extract at doses of 120 mg [22], 240 mg [23], and 300 mg [9] has shown an improvement in sleep quality among healthy participants. Moreover, preclinical studies on *Rhodiola* [24], *Ocimum sanctum* [25], and *Schisandra* [26] extracts did not report any adverse events, indicating their safety. Therefore, in the current study, participants consumed 350-mg capsules of VL-G-A57 or VL-G-E12 twice daily. The placebo capsule contained 350 mg of olive oil. A “double-dummy” design was used to allow IPs to be taken on different schedules. Participants in the VL-G-A57 arm consumed two 350-mg IP capsules in the morning and two placebo capsules at night. Participants in the VL-G-E12 arm consumed one 350-mg IP capsule with one placebo capsule

in the morning and at night. The placebo arm consumed two placebo capsules in the morning and at night. All participants were instructed to consume capsules with food for 60 days.

Participants having at least 90% compliance, assessed through capsule counts and IP diary entries, were considered to have “completed intervention.” To preserve the blinding, the study products and the placebo capsules were matched in size, shape, color, and texture and were packed in identical packaging. The sponsor supplied the products for the study. The products were manufactured in compliance with good manufacturing practices and applicable regulations.

Outcomes

The primary outcome of the study was to assess the change in the stress levels over 60 days of intervention compared to the placebo, and this was measured using the Perceived Stress Scale (PSS). The secondary outcomes of the study were changes in sleep quality, restorative sleep, mental alertness, fatigue, and depression and anxiety. These were assessed using the Pittsburgh Sleep Quality Index (PSQI), Restorative Sleep Questionnaire–Weekly (RSQ-W), Mental Alertness Likert Scale, Fatigue Severity Scale (FSS), and Depression, Anxiety, and Stress Scale-21 (DASS-21) on days 0, 30, and 60, respectively. The participants were instructed to maintain a diet diary and sleep diary. The safety of the intervention was evaluated by measuring vitals (pulse rate, blood pressure) and the occurrence of adverse events throughout the study.

Statistical analysis

A priori power analysis was performed to estimate the required sample size. For the current study, we assumed a large and clinically meaningful difference between groups (Cohen’s $d=0.8$, corresponding to Cohen’s $f\approx0.40$). Using a significance level of $\alpha=0.05$ and 80% power, the calculated sample size was approximately 62 participants per arm, i.e., a total of 186 participants. The primary null hypothesis was that supplementation with VL-G-A57 and VL-G-E12 does not improve the perceived stress as

Table 1 Investigational products composition

VL-G-A57	VL-G-E12
Proprietary blend [hydroethanol extract of <i>Rhodiola rosea</i> root (5% rosavins), milky oat seed (<i>Avena sativa</i>), holy basil (<i>Ocimum sanctum</i>) leaf (4.5% eugenol), and <i>Schisandra chinensis</i> berry, ashwagandha (<i>Withania somnifera</i>) extract] — 260 mg	Proprietary blend [hydroethanol extract of ashwagandha (<i>Withania somnifera</i>) root, ashwagandha (<i>Withania somnifera</i>) root extract] — 175 mg
Excipient blend (vegetable glycerin, water, olive oil)	Excipient blend (vegetable glycerin, water)
Hypromellose capsule	Hypromellose capsule
Total filled weight/capsule — 350 mg	

assessed by PSS (Perceived Stress Scale) in comparison to placebo. For the rejection of the null hypothesis, a two-sided α level of 0.05 was considered significant to indicate the difference observed between IP and placebo. Data was checked for normality using visual assessment. In case of sharp deviation from normality through visual assessment, the distribution of the data was further checked using the Shapiro–Wilk/Kolmogorov–Smirnov test. The analyses were performed using observed cases (OC) only.

The mean change in PSS score on days 30 and 60 from baseline was compared between the IP and the placebo using analysis of covariance (ANCOVA) with treatment as a factor and baseline assessment as a covariate. All efficacy analyses were performed on the full analysis set (FAS) population. To determine which groups were statistically significant compared with the placebo, Dunnett's post hoc test was used. A paired *t*-test was used to assess the within-group comparison from baseline to post-baseline assessment (days 30 and 60). Mean change in the PSQI, RSQ-W, FSS, FSS–VAS, and DASS-21 scores at days 30 and 60 from baseline was assessed similar to the primary endpoints. Data analyses were performed using the R/R Foundation for Statistical

Computing, Vienna, Austria (<https://www.R-project.org/version4.0.5>), and the XLSTAT statistical and data analysis solution, New York, USA (<https://www.xlstat.com/version2021.3.1>).

Results

A total of 186 participants were randomized in the study to one of the three study arms — VL-G-A57, VL-G-E12, and placebo. Of these, 172 participants completed the study. The detailed participant disposition is provided in Fig. 1 below.

Demographics and baseline characteristics

The demographics and baseline characteristics of all the randomized participants are summarized in Table 2. The enrolled participants were predominantly female (67%) with a mean age of 37 years and a mean BMI of 25.30 kg/m². All three study arms were comparable with respect to demographic characteristics.

Perceived stress

Highly stressed individuals, as indicated by PSS scores between 27 and 40, were enrolled in the study. A

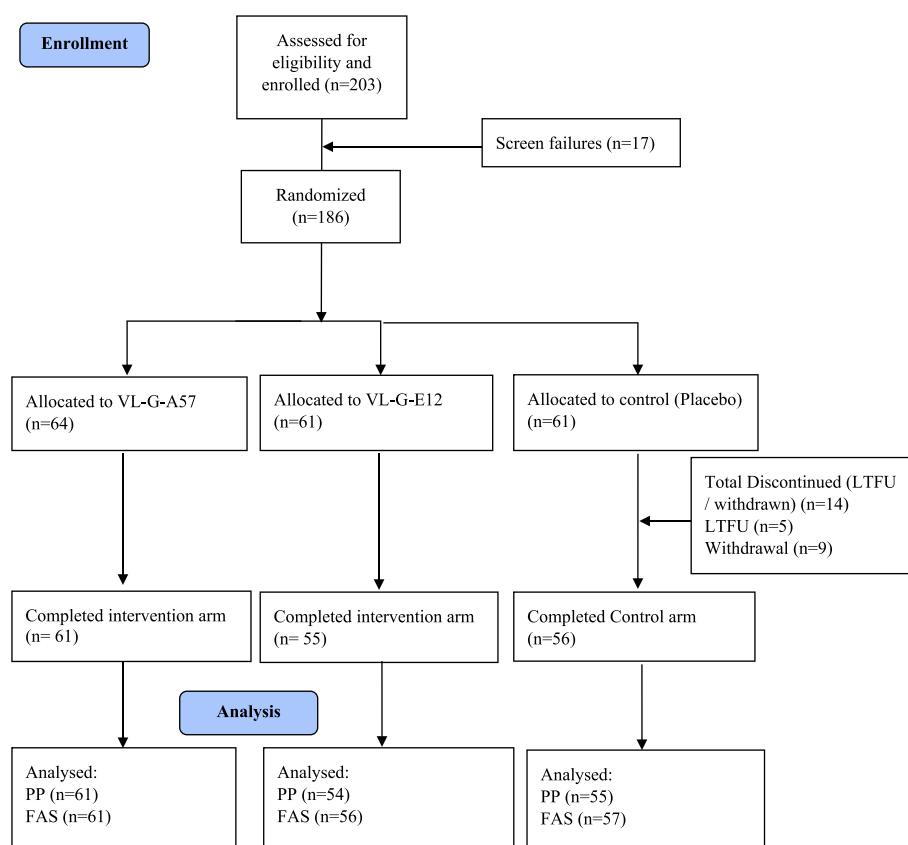


Fig. 1 CONSORT flow chart. Abbreviations: FAS, full analysis set; LTFU, lost to follow-up; PP, per protocol

Table 2 Demographics and baseline characteristics

Parameters	Categories	VL-G-A57 (N=64)	VL-G-E12 (N=61)	Placebo (N=61)	Overall (N=186)	p-value
Age (years)	Mean (SD)	36.66 (7.96)	38.34 (9.41)	37.03 (8.45)	37.33 (8.60)	0.5210 (A)
Gender	Male	18 (28.13%)	21 (34.43%)	22 (36.07%)	61 (32.80%)	0.6056 (C)
	Female	46 (71.88%)	40 (65.57%)	39 (63.93%)	125 (67.20%)	
BMI (kg/m ²)	Mean (SD)	25.16 (2.67)	25.28 (2.38)	25.45 (2.36)	25.30 (2.47)	0.8091 (A)

Notes: For continuous variables, *p*-values were calculated using ANOVA. For categorical variables, *p*-value was calculated using chi-square test

Abbreviations: BMI Body Mass Index, N Number of Participants

significant stress reduction was seen in the IP groups on days 30 and 60, as evident in Table 3. At the end of the study, the two IP groups demonstrated a mean reduction in PSS scores almost twice that of the placebo. This change was also statistically significant ($p < 0.05$) by day 30 for the VL-G-E12 arm and in both the IP arms, VL-G-A57 and VL-G-E12, when compared to placebo ($p < 0.0001$) at day 60 (Table 3).

At day 30, 54.1% and 57.1% of participants moved to the moderate stress level category from high at baseline in the VL-G-A57 and VL-G-E12 groups, respectively, whereas in the placebo group this percentage was only 31.6%. Furthermore, at the end of the study, in the two IP arms, VL-G-A57 and VL-G-E12, only 13.1% and 8.9% of participants had high PSS scores, whereas in the placebo group 56.1% participants remained in the high-stress level category. In the IP arms VL-G-A57 and VL-G-E12, 85.2% and 89.3% of participants, respectively, shifted to the moderate category at the end of the study, and 1.6% participants shifted to the low-stress category in the VL-G-A57 arm, thereby indicating a significant impact of the IPs on stress reduction (Fig. 2).

Fatigue

By the end of the study, both investigational product groups demonstrated greater reductions in fatigue severity compared with placebo. Mean Fatigue Severity Scale (FSS) scores decreased more markedly in the VL-G-A57 and VL-G-E12 groups than in the placebo group. The change was statistically significant between VL-G-A57 and placebo ($p = 0.0003$) and between VL-G-E12 and placebo ($p = 0.0010$) at day 60, as well as across the three study arms ($p = 0.0002$). Also, a statistically significant reduction in the mean FSS–VAS scores was noted at day 30 in the VL-G-E12 arm ($p = 0.0227$). At day 60, compared with placebo, the two IP arms, VL-G-A57 and VL-G-E12, demonstrated statistically significant changes with $p = 0.0362$ and $p = 0.0002$, respectively (Table 4).

The minimal clinically important difference (MCID) for FSS was considered to be a change from baseline of ≤ -1.2 . Participants eliciting this change were considered responders. At the end of the study, the percentage of responders was twice as high in both the IP arms

compared with the placebo (Fig. 3), thereby indicating that the investigational products VL-G-A57 and VL-G-E12 were effective in reducing fatigue over 60 days.

Sleep quality

When compared to placebo, the investigational products demonstrated a statistically significant reduction of scores in the subjective sleep quality, sleep latency, and daytime dysfunction (VL-G-E12: $p = 0.0231$) domains. Additionally, the global PSQI score results were statistically significant in the IP arms when compared to placebo. There was a non-significant change in the score for sleep medication usage as sedatives were prohibited during the study. The change in sleep duration and sleep disturbances score reduced; however, it did not show statistical significance compared to placebo, as participants with intrinsic sleep problems were excluded from the study. Additionally, the global PSQI score results were also statistically significant in the IP arms when compared to placebo (VL-G-A57: $p = 0.0008$, VL-G-E12: $p < 0.0001$) (Table 5).

An improvement in the global PSQI score of ≥ 3 points is considered an MCID, and hence, participants with a ≥ 3 -point reduction at days 30 and 60 were defined as responders. At day 30, VL-G-E12 showed the highest number of responders, 14 (25%). At day 60, as compared to placebo, the two IP arms, VL-G-A57 and VL-G-E12, had a greater number of responders, 32 (52%) and 30 (53%), respectively (Fig. 4).

Restorative sleep

At day 60, both investigational product (IP) groups demonstrated greater improvements in RSQ-W scores compared with placebo. The magnitude of improvement in the IP arms was approximately twofold higher than that observed in the placebo group, and the between-group differences were statistically significant ($p < 0.0001$) (Table 6).

Mental alertness

At day 60, both investigational product (IP) arms showed a numerically greater reduction in mean Mental

Table 3 Perceived Stress Scale — FAS population

Parameters	Categories	VL-G-A57 (N = 61)		VL-G-E12 (N = 56)		Placebo (N = 57)		§p-value
		Mean (SD)	95% CI	Mean (SD)	95% CI	Mean (SD)	95% CI	
PSS scores	Day 0	30.08 (2.44)	(29.46, 30.71)	30.23 (2.33)	(29.61, 30.86)	30.49 (2.27)	(29.89, 31.09)	
	Day 30	26.61 (3.79)	(25.64, 27.58)	25.64 (3.91)	(24.59, 26.69)	27.67 (3.74)	(26.67, 28.66)	
	Change from baseline at day 30	−3.48 (4.02)	(−4.50, −2.45)	−4.59 (3.91)	(−5.64, −3.54)	−2.82 (4.28)	(−3.96, −1.69)	0.0248
	Effect size (Cohen's d)	0.1589		0.4317				
	LS means (95% CI)	−3.60 (−4.55, −2.65)		−4.61 (−5.60, −3.62)		−2.67 (−3.65, −1.69)		
	Difference estimate (95% CI) vs. placebo	−0.93 (−2.48, 0.61)		−1.94 (−3.52, −0.37)				
	* p-value vs placebo	0.3035 (T)		0.0127 (T)				
	Day 60	22.25 (3.56)	(21.33, 23.16)	21.33 (3.72)	(20.32, 22.33)	26.09 (5.06)	(24.73, 27.45)	
	Change from baseline at day 60	−7.84 (4.09)	(−8.88, −6.79)	−8.95 (4.03)	(−10.03, −7.86)	−4.39 (5.52)	(−5.87, −2.91)	< 0.0001
	Effect size (Cohen's d)	0.7101		0.9435				
	LS means (95% CI)	−8.00 (−9.05, −6.95)		−8.95 (−10.05, −7.84)		−4.22 (−5.31, −3.12)		
	Difference estimate (95% CI) vs. placebo	−3.78 (−5.50, −2.06)		−4.73 (−6.49, −2.97)				
	* p-value vs placebo	< 0.0001 (T)		< 0.0001 (T)				

Abbreviations: CI confidence interval, n number of participants, PSS Perceived Stress Scale, SD standard deviations
§ p-values were calculated using ANCOVA with treatment and visit as factor and baseline as covariate
* p-values were calculated using ANCOVA with treatment and visit as factor and baseline as covariate vs. placebo (Dunnett's adjustment)

Alertness Scale scores compared with placebo, suggesting an improvement in mental alertness; however, these differences did not reach statistical significance (Table 7).

Depression and anxiety

A reduction in DASS-21 depression scores was observed in the investigational product arms over the study period. This change was statistically significant in the VL-G-A57 group when compared to placebo ($p=0.0454$) at day 60. The mean scores of the depression subscale indicate that the severity changed from mild at baseline to normal at the end of the study only in the two IP arms, as evident in Table 8. The anxiety subscale scores also showed a statistically significant reduction in the IP arms at day 60 when compared to placebo (VL-G-A57: $p=0.0004$, VL-G-E12: $p=0.0015$). Furthermore, significant reductions in the stress subscale scores were noted in the VL-G-A57 and VL-G-E12 groups, respectively. At the end of the study, the severity in these two IP groups changed to moderate from severe at baseline, whereas the participants in the placebo group remained severely stressed with a mean reduction of only 1.61 (5.52) score. The magnitude of this change was statistically significant when compared to placebo in both the IP arms ($p<0.0001$) (Table 8). These findings are consistent with and further support the efficacy of the investigational products in relation to the study’s primary outcome as assessed by the PSS.

Safety and adverse events

All participants were actively monitored for adverse events at each visit throughout the study. Adverse events were recorded through spontaneous participant reporting as well as investigator-led questioning and were classified based on severity (mild, moderate, severe) and on their relationship to the investigational products. A total of 11 adverse events were recorded during the study, which included headaches (5 events), coughs and colds (2 events each), a hand burn, and a knife cut (1 event each). All adverse events were mild, assessed by the investigators as unrelated to the study intervention, and were resolved in a timely manner without any sequelae. No serious adverse events were reported, and no participant discontinued the study due to adverse events.

Additionally, all the vital parameters (blood pressure and pulse rate) remained within normal limits across all study visits, and no clinically significant changes were observed, supporting the overall safety and tolerability of the investigational products.

Discussion

The “fight-or-flight” response of the adrenal gland is primarily governed by corticotrophic hormones. Adaptogens are agents that stabilize such responses and consequently exhibit neuroprotective activity. Various adaptogens, primarily of herbal origin, have recently gained substantial attention owing to their safety and effectiveness [27]. The present study aimed to study the adaptogenic potential of two such nutritional supplement products — Adrenal Health Daily Support/Daytime HPA (VL-G-A57

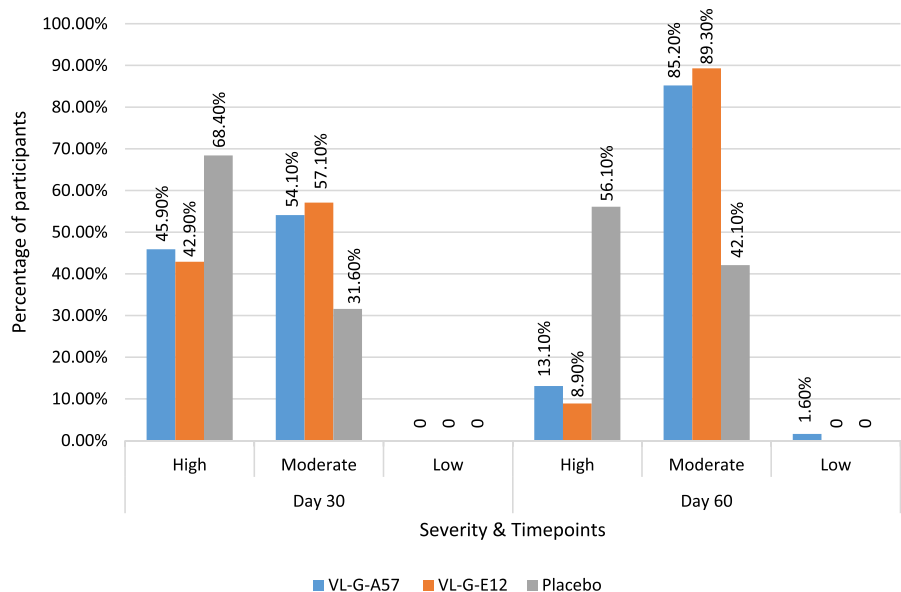


Fig. 2 PSS-severity. Abbreviation: PSS, Perceived Stress Scale

Table 4 Fatigue Severity Scale — FAS population

Parameters	Categories	VL-G-A57 (N = 61)		VL-G-E12 (N = 56)		Placebo (N = 57)		§p-value
		Mean (SD)	95% CI	Mean (SD)	95% CI	Mean (SD)	95% CI	
FSS scores	Day 0	5.16 (0.85)	(4.94, 5.38)	5.14 (1.02)	(4.87, 5.42)	5.18 (0.83)	(4.96, 5.40)	
	Day 30	4.94 (0.86)	(4.72, 5.16)	5.02 (1.00)	(4.75, 5.29)	5.14 (0.87)	(4.91, 5.37)	
	Change from baseline at day 30	−0.22 (0.76)	(−0.41, −0.02)	−0.12 (0.99)	(−0.39, 0.15)	−0.04 (0.57)	(−0.19, 0.12)	0.3625
	LS means (95% CI)	−0.22 (−0.40, −0.04)		−0.13 (−0.31, 0.06)		−0.03 (−0.22, 0.16)		
	Difference estimate (95% CI) vs. placebo	−0.19 (−0.48, 0.11)		−0.10 (−0.40, 0.21)				
	*p-value vs placebo	0.2632 (T)		0.6991 (T)				
	Day 60	4.50 (0.88)	(4.27, 4.72)	4.53 (0.87)	(4.29, 4.76)	5.01 (0.93)	(4.76, 5.26)	
	Change from baseline at day 60	−0.66 (0.67)	(−0.83, −0.49)	−0.63 (0.96)	(−0.89, −0.37)	−0.18 (0.63)	(−0.35, −0.01)	0.0002
	LS means (95% CI)	−0.66 (−0.84, −0.49)		−0.63 (−0.82, −0.45)		−0.17 (−0.35, 0.01)		
	Difference estimate (95% CI) vs. placebo	−0.49 (−0.78, −0.21)		−0.46 (−0.75, −0.17)				
FSS–VAS scores	*p-value vs placebo	0.0003 (T)		0.0010 (T)				
	Day 0	6.67 (1.51)	(6.28, 7.06)	6.84 (1.64)	(6.40, 7.28)	6.70 (1.34)	(6.35, 7.06)	
	Day 30	6.05 (1.15)	(5.76, 6.34)	5.96 (1.61)	(5.53, 6.39)	6.32 (1.10)	(6.02, 6.61)	
	Change from baseline at day 30	−0.62 (1.10)	(−0.90, −0.34)	−0.88 (0.97)	(−1.14, −0.61)	−0.39 (1.15)	(−0.69, −0.08)	0.0414
	LS means (95% CI)	−0.65 (−0.88, −0.42)		−0.84 (−1.08, −0.59)		−0.40 (−0.64, −0.16)		
	Difference estimate (95% CI) vs. placebo	−0.25 (−0.62, 0.13)		−0.44 (−0.82, −0.05)				
	*p-value vs placebo	0.2429 (T)		0.0227 (T)				
	Day 60	5.25 (1.42)	(4.88, 5.61)	5.00 (1.37)	(4.63, 5.37)	5.77 (1.56)	(5.35, 6.19)	
	Change from baseline at day 60	−1.43 (1.27)	(−1.75, −1.10)	−1.87 (1.28)	(−2.22, −1.53)	−0.95 (1.29)	(−1.29, −0.60)	0.0004
	LS means (95% CI)	−1.46 (−1.74, −1.17)		−1.82 (−2.13, −1.52)		−0.96 (−1.26, −0.66)		
	Difference estimate (95% CI) vs. placebo	−0.50 (−0.97, −0.03)		−0.86 (−1.35, −0.38)				
	*p-value vs placebo	0.0362 (T)		0.0002 (T)				

Abbreviations: FSS Fatigue Severity Scale, CI confidence interval, n number of participants, SD standard deviation, VAS Visual Analog Scale

§ p-values were calculated using ANCOVA with treatment and visit as factor and baseline as covariate

* p-values were calculated using ANCOVA with treatment and visit as factor and baseline as covariate vs. placebo (Dunnett's adjustment)

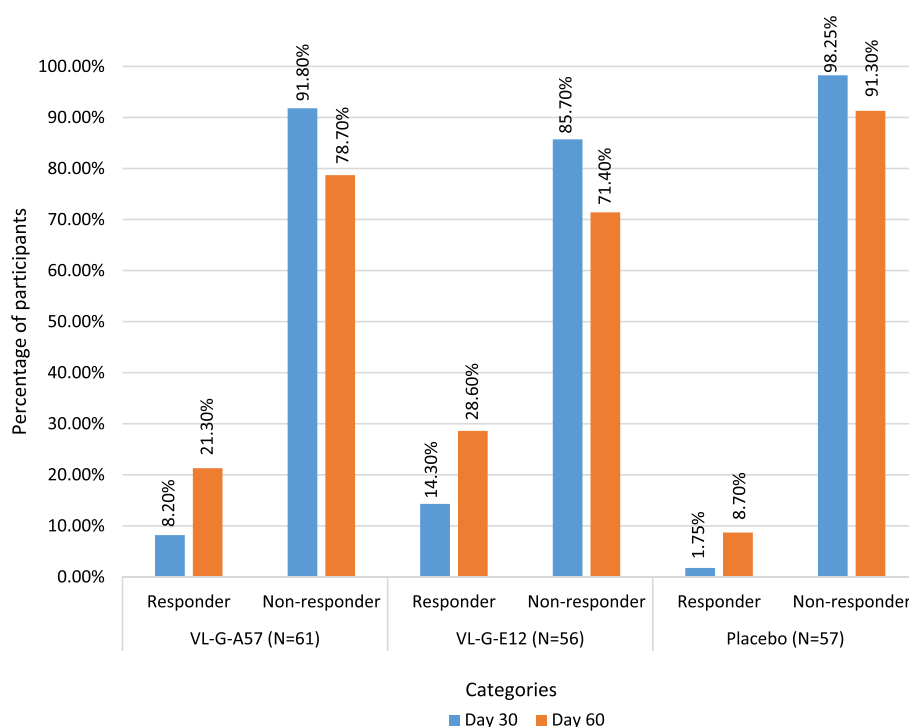


Fig. 3 FSS-responder analysis. Abbreviation: FSS, Fatigue Severity Scale

— containing *Rhodiola rosea*, holy basil (*Ocimum sanctum* L.), oats (*Avena sativa*) in the milky stage, ashwagandha (*Withania somnifera* (L.) Dunal) and *Schisandra chinensis*, and ashwagandha root (VL-G-E12) in liquid filled capsule form. The study was able to meet the primary hypothesis of the adaptogenic potential of both investigational products.

Both VL-G-A57 and VL-G-E12 were able to significantly reduce the PSS scores. These findings were in line with the study conducted by Chandrashekar et al. in stressed individuals, where a similar reduction in PSS (Perceived Stress Scale) scores was observed following 2 months' intake of ashwagandha [28]. Although no validated MCID exists for the PSS, prior studies suggest that reductions of 2–3 points may be considered as clinically meaningful [29]. While we did not perform an MCID analysis, the observed change of −7.84 units and −8.95 units in the VL-G-A57 and VL-G-E12, respectively, can be considered in this context, noting that this interpretation is exploratory. Also, another clinical trial has demonstrated that apart from stress, ashwagandha could significantly reduce anxiety and depression [30]. Stress is known to impair sleep function and sleep quality [31]. The improvement in the perceived stress in the present study also led to an improvement in the sleep quality of participants by VL-G-A57 as well as VL-G-E12 with $p=0.0011$ and $p=0.0002$, respectively. The time

taken for an individual to go from being fully awake to sleep (i.e., sleep latency) also showed positive outcomes in the VL-G-A57 and VL-G-E12 groups. Furthermore, an improvement in sleep efficiency and reduction in sleep disturbance were also noted at the end of the study when compared to the baseline in both the interventional groups. *Schisandra chinensis* and ashwagandha are traditionally known to have sedative effects [9]. Non-restorative sleep (NRS) is known to be associated with stress, anxiety, depression, and daytime fatigue [32]. NRS (i.e., feeling that sleep was restless, light, or of poor quality or awakening feeling unrestored or unrefreshed) affects almost 10% of people worldwide, leading to poor sleep quality, as well as physical and cognitive fatigue. Supplementation with a standardized ashwagandha extract for 6 weeks improved the overall quality of sleep by significantly improving the non-restorative sleep condition in healthy participants in a randomized trial evaluating the effects of ashwagandha on sleep quality in adults [23]. In agreement with these findings, in the current study, an improvement in restorative sleep was noted as assessed by the Restorative Sleep Questionnaire. Participants randomized in the IP arms showed a statistically significant change at day 60 when compared to placebo ($p<0.0001$). The results for the present study support a previous study conducted on this specific ashwagandha root formulation (VL-G-E12), wherein it was concluded that ashwagandha

Table 5 Pittsburgh Sleep Quality Index — FAS population

Parameters	Categories	VL-G-A57 (N=61)		VL-G-E12 (N=56)		Placebo (N=57)		§p-value
		Mean (SD)	95% CI	Mean (SD)	95% CI	Mean (SD)	95% CI	
Subjective sleep quality scores	Day 0	2.18 (0.59)	(2.03, 2.33)	2.21 (0.65)	(2.04, 2.39)	2.19 (0.64)	(2.02, 2.36)	
	Day 30	1.85 (0.51)	(1.72, 1.98)	1.73 (0.62)	(1.57, 1.90)	1.93 (0.62)	(1.76, 2.10)	
	Change from baseline at day 30	-0.33 (0.60)	(-0.48, -0.17)	-0.48 (0.63)	(-0.65, -0.31)	-0.26 (0.52)	(-0.40, -0.13)	0.0781
	LS means (95% CI)	-0.34 (-0.46, -0.21)		-0.47 (-0.60, -0.34)		-0.26 (-0.39, -0.14)		
	Difference estimate (95% CI) vs. placebo	-0.07 (-0.27, 0.13)		-0.21 (-0.42, -0.00)				
	*p-value vs placebo	0.6515 (T)		0.0490 (T)				
	Day 60	1.33 (0.51)	(1.20, 1.46)	1.27 (0.53)	(1.13, 1.41)	1.70 (0.66)	(1.52, 1.87)	
Sleep latency scores	Change from baseline at day 60	-0.85 (0.75)	(-1.04, -0.66)	-0.95 (0.80)	(-1.16, -0.73)	-0.50 (0.81)	(-0.72, -0.28)	0.0001
	LS means (95% CI)	-0.87 (-1.01, -0.73)		-0.93 (-1.08, -0.78)		-0.50 (-0.65, -0.35)		
	Difference estimate (95% CI) vs. placebo	-0.37 (-0.60, -0.13)		-0.43 (-0.66, -0.19)				
	*p-value vs placebo	0.0011 (T)		0.0002 (T)				
	Day 0	2.33 (0.57)	(2.18, 2.47)	2.43 (0.57)	(2.28, 2.58)	2.39 (0.59)	(2.23, 2.54)	
	Day 30	2.20 (0.51)	(2.07, 2.33)	2.09 (0.58)	(1.93, 2.24)	2.25 (0.63)	(2.08, 2.41)	
	Change from baseline at day 30	-0.13 (0.59)	(-0.28, 0.02)	-0.34 (0.61)	(-0.50, -0.18)	-0.14 (0.58)	(-0.29, 0.01)	0.1380
	LS means (95% CI)	-0.16 (-0.29, -0.03)		-0.31 (-0.45, -0.18)		-0.14 (-0.27, -0.00)		
	Difference estimate (95% CI) vs. placebo	-0.02 (-0.23, 0.19)		-0.18 (-0.39, 0.04)				
	*p-value vs placebo	0.9618 (T)		0.1222 (T)				
	Day 60	1.72 (0.66)	(1.55, 1.89)	1.67 (0.64)	(1.50, 1.85)	2.29 (0.68)	(2.10, 2.47)	
	Change from baseline at day 60	-0.61 (0.78)	(-0.81, -0.41)	-0.75 (0.75)	(-0.95, -0.54)	-0.11 (0.87)	(-0.34, 0.13)	<0.0001
	LS means (95% CI)	-0.65 (-0.81, -0.48)		-0.71 (-0.89, -0.54)		-0.10 (-0.27, 0.08)		
	Difference estimate (95% CI) vs. placebo	-0.55 (-0.82, -0.28)		-0.62 (-0.89, -0.34)				
	*p-value vs placebo	<0.0001 (T)		<0.0001 (T)				

Table 5 (continued)

Parameters	Categories	VL-G-A57 (N=61)		VL-G-E12 (N=56)		Placebo (N=57)		[§] p-value
		Mean (SD)	95% CI	Mean (SD)	95% CI	Mean (SD)	95% CI	
Sleep duration scores	Day 0	1.23 (0.62)	(1.07,1.39)	1.25 (0.61)	(1.09, 1.41)	1.19 (0.72)	(1.00, 1.38)	
	Day 30	1.21 (0.61)	(1.06, 1.37)	1.16 (0.56)	(1.01, 1.31)	1.12 (0.71)	(0.93, 1.31)	
	Change from base-line at day 30	-0.02 (0.65)	(-0.18, 0.15)	-0.09 (0.55)	(-0.24, 0.06)	-0.07 (0.42)	(-0.18, 0.04)	0.6936
	LS means (95% CI)	-0.01 (-0.14, 0.11)		-0.08 (-0.21, 0.05)		-0.08 (-0.21, 0.05)		
	Difference estimate (95% CI) vs. placebo	0.07 (-0.13, 0.27)		0.00 (-0.20, 0.21)				
	*p-value vs placebo	0.6695 (T)		0.9993 (T)				
Sleep efficiency scores	Day 60	1.07 (0.48)	(0.94, 1.19)	0.91 (0.44)	(0.79, 1.03)	0.95 (0.44)	(0.83, 1.07)	
	Change from base-line at day 60	-0.16 (0.64)	(-0.33, -0.00)	-0.35 (0.58)	(-0.50, -0.19)	-0.25 (0.58)	(-0.41, -0.09)	0.0877
	LS means (95% CI)	-0.16 (-0.26, -0.06)		-0.33 (-0.43, -0.22)		-0.27 (-0.38, -0.16)		
	Difference estimate (95% CI) vs. placebo	0.11 (-0.06, 0.28)		-0.06 (-0.23, 0.12)				
	*p-value vs placebo	0.2571 (T)		0.6903 (T)				
	Day 0	0.56 (0.90)	(0.33, 0.79)	0.45 (0.74)	(0.25, 0.64)	0.40 (0.70)	(0.22, 0.59)	
Sleep efficiency scores	Day 30	0.41 (0.74)	(0.22, 0.60)	0.36 (0.67)	(0.18, 0.54)	0.40 (0.68)	(0.22, 0.58)	
	Change from base-line at day 30	-0.15 (0.75)	(-0.34, 0.04)	-0.09 (0.51)	(-0.23, 0.05)	0.00 (0.65)	(-0.17, 0.17)	0.6996
	LS means (95% CI)	-0.11 (-0.25, 0.03)		-0.10 (-0.24, 0.04)		-0.03 (-0.17, 0.11)		
	Difference estimate (95% CI) vs. placebo	-0.08 (-0.30, 0.15)		-0.07 (-0.30, 0.16)				
	*p-value vs placebo	0.6526 (T)		0.7189 (T)				
	Day 60	0.13 (0.39)	(0.03, 0.23)	0.09 (0.29)	(0.01, 0.17)	0.14 (0.35)	(0.05, 0.24)	
Sleep efficiency scores	Change from base-line at day 60	-0.43 (0.85)	(-0.64, -0.21)	-0.36 (0.68)	(-0.55, -0.18)	-0.27 (0.62)	(-0.43, -0.10)	0.6051
	LS means (95% CI)	-0.36 (-0.44, -0.28)		-0.38 (-0.47, -0.30)		-0.32 (-0.41, -0.24)		
	Difference estimate (95% CI) vs. placebo	-0.04 (-0.17, 0.09)		-0.06 (-0.19, 0.07)				
	*p-value vs placebo	0.7440 (T)		0.5083 (T)				

Table 5 (continued)

Parameters	Categories	VL-G-A57 (N=61)		VL-G-E12 (N=56)		Placebo (N=57)		[§] p-value
		Mean (SD)	95% CI	Mean (SD)	95% CI	Mean (SD)	95% CI	
Sleep disturbance scores	Day 0	1.62 (0.52)	(1.49, 1.76)	1.63 (0.52)	(1.48, 1.77)	1.61 (0.56)	(1.47, 1.76)	
	Day 30	1.54 (0.50)	(1.41, 1.67)	1.57 (0.57)	(1.42, 1.72)	1.54 (0.50)	(1.41, 1.68)	
	Change from baseline at day 30	-0.08 (0.42)	(-0.19, 0.03)	-0.05 (0.40)	(-0.16, 0.05)	-0.07 (0.32)	(-0.15, 0.01)	0.9045
	LS means (95% CI)	-0.08 (-0.17, 0.01)		-0.05 (-0.15, 0.04)		-0.07 (-0.16, 0.02)		
	Difference estimate (95% CI) vs. placebo	-0.01 (-0.16, 0.14)		0.02 (-0.13, 0.17)				
	*p-value vs placebo	0.9851 (T)		0.9385 (T)				
Use of sleep medication scores	Day 60	1.34 (0.48)	(1.22, 1.47)	1.35 (0.48)	(1.22, 1.48)	1.50 (0.54)	(1.36, 1.64)	
	Change from baseline at day 60	-0.28 (0.52)	(-0.41, -0.15)	-0.27 (0.49)	(-0.40, -0.14)	-0.13 (0.43)	(-0.24, -0.01)	0.0777
	LS means (95% CI)	-0.28 (-0.38, -0.17)		-0.27 (-0.38, -0.16)		-0.12 (-0.23, -0.01)		
	Difference estimate (95% CI) vs. placebo	-0.15 (-0.32, 0.02)		-0.15 (-0.33, 0.02)				
	*p-value vs placebo	0.0806 (T)		0.1002 (T)				
	Day 0	0.07 (0.36)	(-0.03, 0.16)	0.11 (0.49)	(-0.02, 0.24)	0.05 (0.40)	(-0.05, 0.16)	
	Day 30	0.15 (0.51)	(0.02, 0.28)	0.09 (0.39)	(-0.02, 0.19)	0.11 (0.45)	(-0.01, 0.22)	
	Change from baseline at day 30	0.08 (0.64)	(-0.08, 0.25)	-0.02 (0.45)	(-0.14, 0.10)	0.05 (0.61)	(-0.11, 0.21)	0.7379
	LS means (95% CI)	0.07 (-0.04, 0.19)		0.01 (-0.11, 0.13)		0.03 (-0.08, 0.15)		
	Difference estimate (95% CI) vs. placebo	0.04 (-0.15, 0.23)		-0.02 (-0.21, 0.17)				
	*p-value vs placebo	0.8446 (T)		0.9424 (T)				
	Day 60	0.02 (0.13)	(-0.02, 0.05)	0.02 (0.13)	(-0.02, 0.05)	0.09 (0.35)	(-0.00, 0.18)	
	Change from baseline at day 60	-0.05 (0.38)	(-0.15, 0.05)	-0.09 (0.52)	(-0.23, 0.05)	0.04 (0.54)	(-0.11, 0.18)	0.1516
	LS means (95% CI)	-0.06 (-0.12, -0.00)		-0.06 (-0.12, -0.00)		0.01 (-0.05, 0.07)		
	Difference estimate (95% CI) vs. placebo	-0.07 (-0.17, 0.02)		-0.07 (-0.17, 0.03)				
	*p-value vs placebo	0.1461 (T)		0.1787 (T)				

Table 5 (continued)

Parameters	Categories	VL-G-A57 (N=61)		VL-G-E12 (N=56)		Placebo (N=57)		[§] p-value
		Mean (SD)	95% CI	Mean (SD)	95% CI	Mean (SD)	95% CI	
Daytime dysfunction scores	Day 0	1.95 (0.74)	(1.76, 2.14)	2.04 (0.76)	(1.83, 2.24)	1.95 (0.77)	(1.74, 2.15)	
	Day 30	1.74 (0.63)	(1.58, 1.90)	1.82 (0.61)	(1.66, 1.98)	1.84 (0.65)	(1.67, 2.01)	
	Change from baseline at day 30	-0.21 (0.55)	(-0.35, -0.07)	-0.21 (0.49)	(-0.35, -0.08)	-0.11 (0.67)	(-0.28, 0.07)	0.4642
	LS means (95% CI)	-0.22 (-0.34, -0.11)		-0.19 (-0.31, -0.06)		-0.12 (-0.24, 0.00)		
	Difference estimate (95% CI) vs. placebo	-0.11 (-0.30, 0.09)		-0.07 (-0.27, 0.13)				
Global PSQI scores	*p-value vs placebo	0.3650 (T)		0.6473 (T)				
	Day 60	1.48 (0.57)	(1.33, 1.62)	1.42 (0.50)	(1.28, 1.55)	1.63 (0.68)	(1.44, 1.81)	
	Change from baseline at day 60	-0.48 (0.65)	(-0.64, -0.31)	-0.62 (0.78)	(-0.83, -0.41)	-0.32 (0.61)	(-0.48, -0.16)	0.0399
	LS means (95% CI)	-0.49 (-0.62, -0.36)		-0.58 (-0.72, -0.45)		-0.34 (-0.47, -0.21)		
	Difference estimate (95% CI) vs. placebo	-0.15 (-0.36, 0.06)		-0.24 (-0.46, -0.03)				
	*p-value vs placebo	0.1849 (T)		0.0231 (T)				
	Day 0	9.93 (2.87)	(9.20, 10.67)	10.11 (2.81)	(9.36, 10.86)	9.79 (2.85)	(9.03, 10.54)	
	Day 30	9.10 (2.44)	(8.47, 9.72)	8.82 (2.40)	(8.18, 9.46)	9.19 (2.59)	(8.51, 9.88)	
	Change from baseline at day 30	-0.84 (1.93)	(-1.33, -0.34)	-1.29 (1.93)	(-1.80, -0.77)	-0.60 (1.85)	(-1.09, -0.11)	0.1620
	LS means (95% CI)	-0.84 (-1.25, -0.43)		-1.23 (-1.66, -0.80)		-0.65 (-1.08, -0.22)		
Daytime dysfunction scores	Difference estimate (95% CI) vs. placebo	-0.19 (-0.86, 0.48)		-0.58 (-1.27, 0.11)				
	*p-value vs placebo	0.7535 (T)		0.1105 (T)				
	Day 60	7.08 (1.97)	(6.58, 7.59)	6.73 (1.88)	(6.22, 7.24)	8.29 (1.97)	(7.76, 8.81)	
	Change from baseline at day 60	-2.85 (2.73)	(-3.55, -2.15)	-3.38 (2.81)	(-4.14, -2.62)	-1.54 (3.11)	(-2.37, -0.70)	<0.0001 (T)
	LS means (95% CI)	-2.87 (-3.33, -2.40)		-3.26 (-3.75, -2.77)		-1.64 (-2.12, -1.15)		
Daytime dysfunction scores	Difference estimate (95% CI) vs. placebo	-1.23 (-1.99, -0.47)		-1.62 (-2.40, -0.84)				
	*p-value vs placebo	0.0008 (T)		<0.0001 (T)				

Abbreviations: CI confidence interval, PSQI Pittsburgh Sleep Quality Index, n number of participants, SD standard deviation

[§] p-values were calculated using ANCOVA with treatment and visit as factor and baseline as covariate

* p-values were calculated using ANCOVA with treatment and visit as factor and baseline as covariate vs. placebo (Dunnett's adjustment)

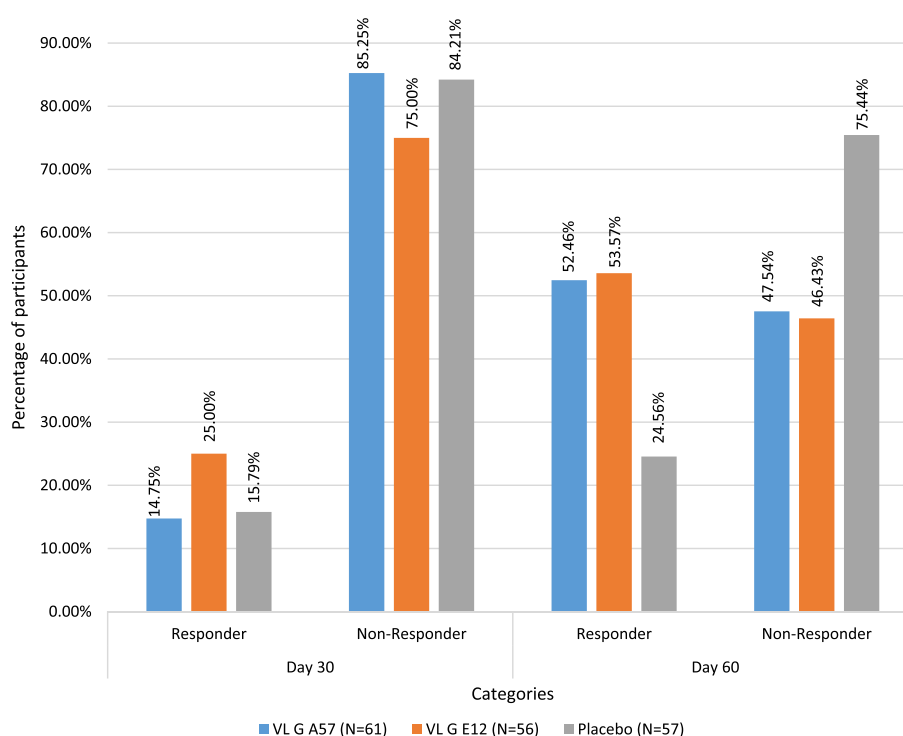


Fig. 4 Global PSQI responder analysis. Abbreviation: PSQI, Pittsburgh Sleep Quality Index

showed a positive impact on the final sleep as well as final stress in adult individuals in 30 days [33]. In the current study, it was noted that the participants in the IP groups experienced a decrease in the severity of sub-clinical depression from mild to normal after 60 days of intervention. Moreover, a statistically significant change was observed for subclinical anxiety in the two IP arms when compared to baseline and placebo [VL-G-A57 ($p=0.0004$) and VL-G-E12 ($p=0.0015$)]. Also, a significant change in the severity of stress was demonstrated by the IP groups from severe at the beginning of the study to moderate at the end of the study. This change in both the IP arms, VL-G-A57 and VL-G-E12, was statistically significant when compared to baseline and placebo ($p<0.0001$). Furthermore, a slight improvement in mental alertness was observed as compared to the baseline in the IP arms VL-G-A57 and VL-G-E12. These results align with the literature; in a randomized, double-blind, placebo-controlled study of 60 participants with insomnia, improved PSQI and DASS-21 scores were reported post-ashwagandha intervention. A greater number of participants with ashwagandha were found mentally alert as compared to placebo at 5 weeks, with a gradual improvement from the baseline to the final assessment noted [9]. In a randomized, double-blind, placebo-controlled clinical trial conducted by Darbinyan et al., individuals with depression were randomized to receive either *R. rosea* or

placebo over 6 weeks. The study concluded that participants in the *R. rosea* group experienced an improvement in overall depression and other mood symptoms [34]. Several clinical trials have shown that adaptogens have the potential to exert an anti-fatigue effect and thereby increase mental work capacity against a stress and fatigue background [7]. In the current study, the IP arms, when compared to placebo, elicited a significant impact in the reduction of fatigue as assessed by FSS scores [VL-G-A57: $p=0.0003$ and VL-G-E12: $p=0.0010$]. Strong scientific evidence is available for *Rhodiola rosea* extract, one of the components of VL-G-A57, which improved attention, cognitive function, and mental performance in fatigue and in chronic fatigue syndrome. Similarly, good scientific evidence has been documented in which *Schisandra chinensis* has increased endurance and mental performance in individuals with mild fatigue and weakness [6].

Often, adaptogens are found to increase the state of nonspecific resistance to stress and decrease sensitivity to stressors, resulting in HPA axis support [7]. Modulation of neuroendocrine signaling represents an additional potential mechanism through which adaptogens may exert stress-reducing effects [35]. A notable aspect of this study is the direct comparison of two distinct formulations, a multicomponent formulation, VL-G-A57, and an ashwagandha extract formulation,

Table 6 Restorative Sleep Questionnaire — FAS population

Parameters	Categories	VL-G-A57 (N = 61)		VL-G-E12 (N = 56)		Placebo (N = 57)		§p-value
		Mean (SD)	95% CI	Mean (SD)	95% CI	Mean (SD)	95% CI	
RSQ-W scores	Day 0	28.87 (8.57)	(26.68, 31.06)	29.86 (8.41)	(27.61, 32.11)	28.80 (9.12)	(26.38, 31.22)	
	Day 30	37.39 (10.18)	(34.78, 39.99)	37.95 (11.44)	(34.88, 41.01)	34.45 (9.72)	(31.88, 37.03)	
	Change from baseline at day 30	8.52 (8.53)	(6.33, 10.70)	8.09 (9.60)	(5.52, 10.66)	5.65 (9.09)	(3.24, 8.07)	0.1316
	LS means (95% CI)	8.42 (6.23, 10.61)		8.31 (6.02, 10.59)		5.54 (3.27, 7.80)		
	Difference estimate (95% CI) vs. placebo	2.88 (−0.67, 6.44)		2.77 (−0.86, 6.41)				
	*p-value vs placebo	0.1283 (T)		0.1597 (T)				
	Day 60	46.17 (11.11)	(43.33, 49.02)	47.22 (9.83)	(44.57, 49.88)	37.80 (12.93)	(34.34, 41.26)	
	Change from baseline at day 60	17.30 (8.99)	(15.00, 19.61)	17.58 (10.46)	(14.75, 20.40)	8.93 (11.78)	(5.77, 12.08)	<0.0001
	LS means (95% CI)	17.21 (14.69, 19.73)		17.77 (15.12, 20.42)		8.84 (6.21, 11.47)		
	Difference estimate (95% CI) vs. placebo	8.38 (4.26, 12.49)		8.93 (4.71, 13.15)				
	*p-value vs placebo	<0.0001 (T)		<0.0001 (T)				

Abbreviations: CI confidence interval, n number of participants, RSQ-W/Restorative Sleep Quality—Weekly version, SD standard deviation
[§] p-values were calculated using ANCOVA with treatment and visit as factor and baseline as covariate
^{*} p-values were calculated using ANCOVA with treatment and visit as factor and baseline as covariate vs. placebo (Dunnett's adjustment)

Table 7 Mental Alertness Scale — FAS population

Parameters	Categories	VL-G-A57 (N = 61)		VL-G-E12 (N = 56)		Placebo (N = 57)		[§] p-value
		Mean (SD)	95% CI	Mean (SD)	95% CI	Mean (SD)	95% CI	
Mental Alertness Scale scores	Day 0	2.33 (0.47)	(2.21, 2.45)	2.32 (0.47)	(2.20, 2.45)	2.26 (0.48)	(2.14, 2.39)	0.4402
	Day 30	2.02 (0.29)	(1.94, 2.09)	1.98 (0.40)	(1.87, 2.09)	2.05 (0.35)	(1.96, 2.15)	
	Change from baseline at day 30	−0.31 (0.50)	(−0.44, −0.18)	−0.34 (0.51)	(−0.48, −0.20)	−0.21 (0.53)	(−0.35, −0.07)	
	LS means (95% CI)	−0.29 (−0.38, −0.21)		−0.33 (−0.41, −0.24)		−0.24 (−0.33, −0.16)		
	Difference estimate (95% CI) vs. placebo	−0.05 (−0.19, 0.09)		−0.08 (−0.22, 0.06)				
	*p-value vs placebo	0.6556 (T)		0.3382 (T)				
	Day 60	1.87 (0.39)	(1.77, 1.97)	1.89 (0.37)	(1.79, 1.99)	2.00 (0.47)	(1.87, 2.13)	
	Change from baseline at day 60	−0.46 (0.53)	(−0.60, −0.32)	−0.44 (0.54)	(−0.58, −0.29)	−0.27 (0.59)	(−0.43, −0.11)	
	LS means (95% CI)	−0.44 (−0.54, −0.34)		−0.42 (−0.53, −0.31)		−0.30 (−0.41, −0.19)		
	Difference estimate (95% CI) vs. placebo	−0.14 (−0.31, 0.02)		−0.12 (−0.29, 0.05)				
	*p-value vs placebo	0.0992 (T)		0.1968 (T)				

Abbreviations: CI confidence interval, n number of participants, SD standard deviation
[§] p-values were calculated using ANCOVA with treatment and visit as factor and baseline as covariate
* p-values were calculated using ANCOVA with treatment and visit as factor and baseline as covariate vs. placebo (Dunnett's adjustment)

Table 8 Depression, Anxiety, and Stress Scale-21 — FAS population

Parameters	Categories	VL-G-A57 (N = 61)		VL-G-E12 (N = 56)		Placebo (N = 57)		^{\$} p-value
		Mean (SD)	95% CI	Mean (SD)	95% CI	Mean (SD)	95% CI	
Depression	Day 0	11.41 (5.72)	(9.95, 12.87)	11.82 (6.36)	(10.12, 13.53)	12.32 (7.64)	(10.29, 14.34)	0.9443
	Day 30	11.05 (7.07)	(9.24, 12.86)	11.14 (5.87)	(9.57, 12.71)	11.33 (6.63)	(9.58, 13.09)	
	Change from baseline at day 30	−0.36 (6.52)	(−2.03, 1.31)	−0.68 (3.86)	(−1.71, 0.35)	−0.98 (5.50)	(−2.44, 0.48)	
	LS means (95% CI)	−0.51 (−1.76, 0.74)		−0.68 (−1.99, 0.62)		−0.82 (−2.11, 0.48)		
	Difference estimate (95% CI) vs. placebo	0.31 (−1.73, 2.34)		0.13 (−1.95, 2.21)				
	*p-value vs placebo	0.9203 (T)		0.9852 (T)				
	Day 60	8.56 (5.37)	(7.18, 9.93)	9.56 (5.67)	(8.03, 11.10)	11.11 (6.58)	(9.34, 12.87)	0.0786
	Change from baseline at day 60	−2.85 (4.80)	(−4.08, −1.62)	−2.15 (5.72)	(−3.69, −0.60)	−1.39 (5.82)	(−2.95, 0.17)	
	LS means (95% CI)	−3.05 (−4.22, −1.88)		−2.21 (−3.44, −0.98)		−1.11 (−2.33, 0.11)		
	Difference estimate (95% CI) vs. placebo	−1.94 (−3.85, −0.03)		−1.10 (−3.06, 0.85)				
	*p-value vs placebo	0.0454 (T)		0.3476 (T)				
Anxiety	Day 0	22.72 (9.91)	(20.18, 25.26)	22.54 (10.19)	(19.81, 25.26)	22.98 (9.43)	(20.48, 25.49)	0.7934
	Day 30	20.82 (9.52)	(18.38, 23.26)	20.57 (9.70)	(17.97, 23.17)	21.54 (9.12)	(19.12, 23.96)	
	Change from baseline at day 30	−1.90 (6.52)	(−3.57, −0.23)	−1.96 (3.92)	(−3.01, −0.92)	−1.44 (5.64)	(−2.94, 0.06)	
	LS means (95% CI)	−1.91 (−3.21, −0.60)		−2.01 (−3.37, −0.64)		−1.39 (−2.74, −0.04)		
	Difference estimate (95% CI) vs. placebo	−0.51 (−2.64, 1.61)		−0.61 (−2.78, 1.55)				
	*p-value vs placebo	0.8102 (T)		0.7523 (T)				
	Day 60	15.70 (7.92)	(13.68, 17.73)	15.93 (8.34)	(13.67, 18.18)	19.54 (7.74)	(17.46, 21.61)	0.0003
	Change from baseline at day 60	−7.02 (5.61)	(−8.45, −5.58)	−6.73 (6.70)	(−8.54, −4.91)	−3.61 (6.50)	(−5.35, −1.87)	
	LS means (95% CI)	−7.06 (−8.35, −5.77)		−6.80 (−8.15, −5.44)		−3.49 (−4.84, −2.15)		
	Difference estimate (95% CI) vs. placebo	−3.57 (−5.67, −1.47)		−3.30 (−5.46, −1.15)				
	*p-value vs placebo	0.0004 (T)		0.0015 (T)				

Table 8 (continued)

Parameters	Categories	VL-G-A57 (N = 61)		VL-G-E12 (N = 56)		Placebo (N = 57)		[§] p-value
		Mean (SD)	95% CI	Mean (SD)	95% CI	Mean (SD)	95% CI	
Stress	Day 0	30.00 (8.23)	(27.89, 32.11)	30.29 (7.95)	(28.16, 32.41)	29.89 (8.35)	(27.68, 32.11)	0.4234
	Day 30	27.77 (8.82)	(25.51, 30.03)	27.36 (8.43)	(25.10, 29.61)	28.46 (8.38)	(26.23, 30.68)	
	Change from baseline at day 30	-2.23 (6.75)	(-3.96, -0.50)	-2.93 (5.04)	(-4.28, -1.58)	-1.44 (5.84)	(-2.99, 0.11)	
	LS means (95% CI)	-2.24 (-3.68, -0.80)		-2.88 (-4.38, -1.38)		-1.47 (-2.96, 0.01)		
	Difference estimate (95% CI) vs. placebo	-0.77 (-3.11, 1.57)		-1.40 (-3.79, 0.98)				
	*p-value vs placebo	0.6836 (T)		0.3194 (T)				
	Day 60	22.30 (9.85)	(19.77, 24.82)	21.20 (8.94)	(18.78, 23.62)	28.46 (8.88)	(26.09, 30.84)	< 0.0001
	Change from baseline at day 60	-7.70 (6.31)	(-9.32, -6.09)	-9.16 (6.05)	(-10.80, -7.53)	-1.61 (5.52)	(-3.09, -0.13)	
	LS means (95% CI)	-7.72 (-9.22, -6.23)		-9.14 (-10.71, -7.56)		-1.62 (-3.17, -0.06)		
	Difference estimate (95% CI) vs. placebo	-6.11 (-8.54, -3.67)		-7.52 (-10.02, -5.02)				
	*p-value vs placebo	< 0.0001 (T)		< 0.0001 (T)				

Abbreviations: CI, confidence interval; n, number of participants; SD, standard deviations

[§] p-values were calculated using ANCOVA with treatment and visit as factor and baseline as covariate

* p-values were calculated using ANCOVA with treatment and visit as factor and baseline as covariate vs. placebo (Dunnett's adjustment)

VL-G-E12, within a single randomized controlled trial. To our knowledge, no prior studies have reported such a direct comparison, making this among the first investigations of its kind. While both investigational products demonstrated improvements in stress and fatigue-related outcomes compared with placebo, differences in the timing and pattern of response were observed. VL-G-E12 was associated with greater improvements in stress and sleep quality, whereas VL-G-A57 showed more sustained effects on mood-related outcomes such as depression and anxiety over the study period. These findings suggest formulation-specific benefits.

The differential effects observed may be attributable to differences in the botanical composition of the two formulations. VL-G-A57 combines adaptogens and nerve tonics that may support longer-term regulation of neuroendocrine pathways [35], whereas VL-G-E12 may exert more immediate effects on sleep quality and stress through modulation of GABAergic neurotransmission

[36]. Thus, HPA axis regulation can be proposed as a plausible mechanism; however, mechanistic interpretations remain speculative. Also, the study was conducted exclusively in India, with a majority of participants being female (~67%). As such, the generalizability of the findings to other populations, including different geographic regions, cultural contexts, or gender distributions, may be limited. Thus, future studies that include physiological markers (ACTH, cortisol, etc.) in more diverse populations are warranted.

The current study demonstrates the stress resistance property of the adrenal support formula; however, the molecular mechanism has not been elucidated.

The investigational products were well tolerated, with 11 adverse events reported. All of them were mild in nature, and none was attributed to the study interventions. Additionally, all vital parameters remained within normal limits throughout the study, further supporting the safety of the investigational products.

Conclusion

The current randomized controlled trial met its primary objective, suggesting that VL-G-A57 and VL-G-E12 may help in reducing stress in otherwise healthy and active adults in 60 days. Additionally, the study also met the secondary outcomes of improving sleep and reducing anxiety, subclinical depression, and fatigue. However, the mechanism of action of these investigational products has not been explained. Both products were found to be safe and well-tolerated.

Abbreviations

ANCOVA	Analysis of Covariance
BMI	Body Mass Index
CNS	Central Nervous System
DASS-21	Depression, Anxiety, and Stress Scale-21
FAS	Full Analysis Set
FSS	Fatigue Severity Scale
HPA	Hypothalamic–Pituitary–Adrenal
IP	Investigational Product
IPAQ-SF	International Physical Activity Questionnaire-Short Form
LTFU	Lost-to-follow-up
MCID	Minimal Clinically Important Difference
NRS	Non-restorative Sleep
PP	Per Protocol
PSQI	Pittsburgh Sleep Quality Index
PSS	Perceived Stress Scale
RSQ-W	Restorative Sleep Questionnaire–Weekly

Acknowledgements

The authors are grateful to all the participants who volunteered for the study. The authors thank Gaia Herbs, USA, for providing the study products and Vedic Lifesciences, India, for facilitating the study.

Authors' contributions

All authors made a significant contribution to the work reported, whether that is in the conception, study design, execution, acquisition of data, analysis, interpretation, or in all these areas; took part in drafting, revising, and critically reviewing the article; gave final approval of the version to be published; have agreed on the journal to which the article has been submitted; and agree to be accountable for all aspects of the work.

Funding

Gaia Herbs provided financial support and investigational products for the trial. The sponsor had no role in the study design, data collection, data management, statistical analysis, or interpretation of the findings. Their involvement was limited to providing product-related information and logistical support. All statistical analyses were performed independently by the contracted biostatistical team, who had full access to the dataset.

Data availability

The data used in the study are available on reasonable request from the corresponding author, with due permission from the sponsor.

Declarations

Ethics approval and consent to participate

The present study was conducted in compliance with the Declaration of Helsinki (Ethical Principles for Medical Research Involving Human Subjects, World Medical Association, General Assembly, Seoul 2008), International Conference on Harmonization–Good Clinical Practice (ICH–GCP) — 2016, and Ethical Guidelines for Biomedical Research on Human Participants, 2006 (Indian Council of Medical Research, India). It was approved by an Independent Ethics Committee, the Harmony Ethical Research Committee (Reg. No.: ECR/1411/Inst/MH/2020), to safeguard the rights, safety, and well-being of all trial participants. The participants who provided voluntary, written informed consent were enrolled in the study. Vedic Lifesciences, Mumbai, India, monitored and

audited the study to ensure the study protocol and ICH–GCP compliance. The study report conformed to the Consolidated Standard Reporting of Trials (CONSORT) guidelines.

Consent for publication

Not applicable.

Competing interests

Erin McKinney and Jeremy Stewart were affiliated with Gaia Herbs during the conduct of the study. Rajesh Kewalramani served as both the principal investigator and as an author of this study, conducted under the supervision of Vedic Lifesciences. The authors declare no competing interests.

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Received: 3 September 2025 Accepted: 22 January 2026

Published online: 09 February 2026

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